

Tableau Data Visualization Assignment Report

Itziar Morales Rodríguez

May 24, 2026

Contents

1	Introduction	2
1.1	Methodology	2
1.2	Sources	2
1.3	AI Usage	3
2	Datasets	3
2.1	Dataset 1: Firearm Suicide Proxy for Household Gun Ownership, 1949–2023	3
2.2	Dataset 2: State Firearm Law Database, 1976–2024	3
2.3	Join Strategy	4
2.4	Analysis	4
2.5	Processing	4
2.6	Research Questions	4
2.6.1	Question 1: Homicide Rates and the Law Landscape	5
2.6.2	Question 2: Homicide Trends by Law Profile Over Time	5
2.6.3	Question 3: State-Level Law History and Homicide Standing	5
3	Visualization Prep	5
3.1	Storytelling	5
3.2	Colors and Themes	5
3.3	Tableau Feature Choices	6
4	Visualizations	6
4.1	Q1 – Homicide Rates and the Law Landscape	7
4.2	Q2 – Homicide Trends by Law Profile Over Time	9
4.3	Q3 – State-Level Law History and Homicide Standing	10
4.4	Dashboards	12
4.5	Visualization Summary	14
5	Conclusions	14
6	References	15

1 Introduction

This project explores the relationship between state firearm legislation and gun violence in the United States using data spanning 1976–2023. While estimated household gun ownership is included as a contextual variable, it is not treated as the primary explanatory factor. Evidence from countries such as Finland and Switzerland where high gun ownership coexists with low rates of gun violence, suggests that the regulatory environment surrounding firearms may matter more than ownership prevalence alone. This project therefore focuses on whether the strength and type of state firearm laws is associated with lower rates of firearm homicide, and whether changes in legislation over time correspond to measurable shifts in a state’s gun violence standing.

1.1 Methodology

The project followed the process below to achieve the goal described above:

1. Dataset selection and join strategy design
2. Data analysis outside Tableau
3. Research questions derived from data exploration
4. Visualization types needed to answer each question and storytell
5. Data source connection, relationships, and calculated fields in Tableau
6. Tableau feature selection (parameters, sets, filters, actions, tooltips)
7. Dashboard design and interactive storytelling
8. Use of AI assistance for tedious but mechanical tasks such as generating calculated fields (e.g. a state-name-to-abbreviation mapping for all 50 states), report structuring, and dataset pairing strategy. All AI output was reviewed critically before use (see Section 1.3)

1.2 Sources

Numerous sources are used throughout the project such as:

- Official dataset documentation and codebooks^{1,2}
- Harvard Dataverse and Tufts CTSI dataset metadata^{3,2}
- Academic literature on firearm ownership proxies^{4,3}
- Academic literature on gun law effects^{5,6,7}
- Tableau official documentation⁸
- Lecture notebooks and course materials
- AI assistance (declared in Section 1.3)

Regardless of the source, these are indicated through bibliography and references, or in the case of AI, an explicit note indicating the model and prompt used.

1.3 AI Usage

AI was used during this project in a critical and supervised manner. All AI generated suggestions were reviewed and scrutinized before being applied, with particular attention to the risk of hallucination. Specifically, AI was used for the following tasks:

- **Dataset research:** identifying and evaluating candidate datasets suitable for a longitudinal U.S. firearm violence analysis, including assessing their grain, coverage, and compatibility for joining
- **Mechanical calculated fields:** generating tedious but deterministic Tableau expressions such as a state-name-to-abbreviation lookup mapping all 50 U.S. states
- **Report structuring:** assistance with LaTeX formatting, section organisation, and bibliography management
- **Supplementary chart generation:** one additional visualization was produced outside of Tableau using AI-assisted code generation, included as a supplementary figure in the relevant section

AI was not used to generate data, draw analytical conclusions, produce Tableau visualizations, or write the interpretation of results.

2 Datasets

Two public datasets were selected and used in conjunction for this project. Both are accessible through their corresponding links and are available in different formats.

2.1 Dataset 1: Firearm Suicide Proxy for Household Gun Ownership, 1949–2023

The first dataset¹ is a state-year level proxy for household gun ownership in the United States. Direct gun ownership data is not publicly available at the state level, so this dataset uses the firearm suicide proxy methodology: the proportion of suicides committed with a firearm (FSS) is used as an indirect estimate of household gun prevalence by state and year. It is hosted on Harvard Dataverse.

- **Source:** Harvard Dataverse, Kang et al.³
- **Grain:** One row per U.S. state per year
- **Key columns:** State, Year, FSS (firearm suicide share), firearm suicides, total suicides, firearm homicide rate, total homicides, population
- **Date span:** 1949–2023 (75 years, 50 states + D.C.)
- **Format:** Comma-separated values (.csv)

This dataset was chosen because it provides a long term, population adjusted, and academically validated measure of gun prevalence that can be compared across states and over time. FSS is not a literal count of guns per household but an indirect proxy widely used in public health and policy research.⁴ FSS is retained in this project as a contextual indicator of estimated gun prevalence on the scatter plot; firearm homicide rate serves as the primary outcome measure throughout.

2.2 Dataset 2: State Firearm Law Database, 1976–2024

The second dataset² is maintained by the State Firearm Law Research Group at Tufts University and tracks the presence or absence of 72 firearm safety law provisions across all 50 U.S. states for every year from 1976 to 2024.

- **Source:** Tufts CTSI — State Firearm Law Database²

- **Grain:** One row per U.S. state per year
- **Key columns:** State, Year, 72 binary law indicator columns (e.g. permit required, background check, waiting period, stand-your-ground, child access prevention)
- **Date span:** 1976–2024
- **Format:** Excel (.xlsx)

This dataset was chosen because it provides a historically accurate, year-by-year record of state firearm legislation, enabling analysis of whether law changes correspond to changes in gun violence rates. It is non-partisan by design and is widely used in academic public health research.²

2.3 Join Strategy

Both datasets were connected in Tableau using a **relationship** on the shared keys `state` and `year`, configured with **many-to-many** cardinality and **some records match** referential integrity on both sides. Many-to-many cardinality is appropriate here because both tables share the same grain (one row per state per year) and neither side acts as a unique dimension table. The “some records match” setting instructs Tableau not to assume all keys are present in both tables, handling non-overlapping rows safely without discarding data prematurely. Tableau resolves the relationship by performing aggregations on each table independently before combining results, preventing row duplication. The overlapping date range of both datasets yields an effective analysis window of **1976–2023**.

2.4 Analysis

Prior to connecting the data in Tableau, both files were inspected to understand structure, null patterns, and label consistency. Key observations:

- Both datasets use identical full state name format (e.g. Alabama, North Carolina), allowing for a join on `state`
- The ownership proxy file includes several pre-calculated rate columns derived from raw counts and population
- The law database contains 72 binary columns but it also includes a total law count column that can be used
- Both datasets include a sufficient overlapping year window between 1976–2023 that can be used to join

2.5 Processing

Both source files were connected to Tableau without modification to the raw data. No column renaming, deduplication, or external pre-processing was required. The only structural adjustment was the exclusion of the District of Columbia via a Tableau filter, as it is present in the ownership proxy dataset but absent from the state firearm laws dataset, which would have prevented a valid join for that entity.

2.6 Research Questions

Three questions were formulated based on data exploration and storytelling intent:

1. Is a state’s firearm homicide rate associated with the number and type of firearm laws it has in place?
2. How have firearm homicide rates evolved over time depending on the firearm laws passed in each state, whether measured by the quantity of laws or by the categories of law types enacted?
3. For a specific state, does passing more firearm laws correspond to a reduction in firearm homicide rates, and how does this affect its ranking relative to other states?

2.6.1 Question 1: Homicide Rates and the Law Landscape

This is the entry point of the analysis. Before drawing any temporal or causal conclusions, it is necessary to establish whether a cross-sectional pattern exists: do states that have enacted more laws or certain types of laws, tend to have lower firearm homicide rates? The question is meant to ask whether the law and violence landscape are associated when viewed together. FSS is included as a contextual variable to show where estimated gun ownership sits in this picture, without treating it as an outcome or a direct cause.

2.6.2 Question 2: Homicide Trends by Law Profile Over Time

If a cross-sectional association exists, the natural follow up is whether it has strengthened or weakened over time. This question was chosen because the datasets span nearly five decades, allowing for more temporal depth. By grouping states according to their law profile (types of laws or number of laws passed), it becomes possible to observe whether the trajectories have diverged, and whether that divergence accelerated after key legislative periods such as the early 1990s.

2.6.3 Question 3: State-Level Law History and Homicide Standing

The first two questions analyze the broader picture whilst this third question brings the analysis down to the individual state, making it personally relevant and explorable for any viewer. The motivation is to let someone examine their own state: when did it pass laws, how many, and does its current homicide ranking reflect that legislative history? It is an exploration due to the nature of each state being individualized, making it harder to draw causality unlike with the broader analysis.

3 Visualization Prep

Before building charts in Tableau, design and structural decisions were made to align visualizations with the storytelling intent and assignment requirements.

3.1 Storytelling

The project follows a four-act narrative structure. Act 1 opens with the geographic baseline: where gun violence is highest and what the law landscape looks like across states. Act 2 introduces the temporal dimension: how homicide rates have diverged between state groups defined by their law profile over time. Act 3 connects both dimensions in the ownership-laws-violence scatter. Act 4 closes with the state-level exploration: does a specific state's legislative history correspond to an improvement in its standing? The viewer moves from observation to exploration to interpretation.

3.2 Colors and Themes

Colors were chosen to encode meaning consistently:

- A **sequential red palette** encodes gun violence rates on maps and scatter plots, where darker shades indicate higher rates.
- A **diverging palette** separates high-law from low-law states in trend and comparison views.
- **Set-based color** encodes user-selected states (highlighted) vs. unselected states (grey) in the interactive bar chart.
- Backgrounds are white and gridlines are minimal to reduce visual noise and keep focus on data marks.

3.3 Tableau Feature Choices

The following Tableau features were planned and used across this project:

- **Calculated fields:** Composite law count score (sum of 72 binary columns), law category classification, above/below national average flag, estimated gun-owning households
- **Parameters:** Year selector, Top N selector for ranked bar chart, State selector, Law-categorization toggle (law quantity tier vs. law type group)
- **Sets and Combined Sets:** Selected states set for interactive bar chart, top/bottom N sets combined for ranked bar chart
- **Filters:** State multiselect, year range
- **Dashboard actions:** Set action (click state on map to highlight it in the bar chart and update the state line chart)
- **Reference lines:** National average in trend views
- **Trend lines:** Linear trend in scatter plot (from Analytics pane)

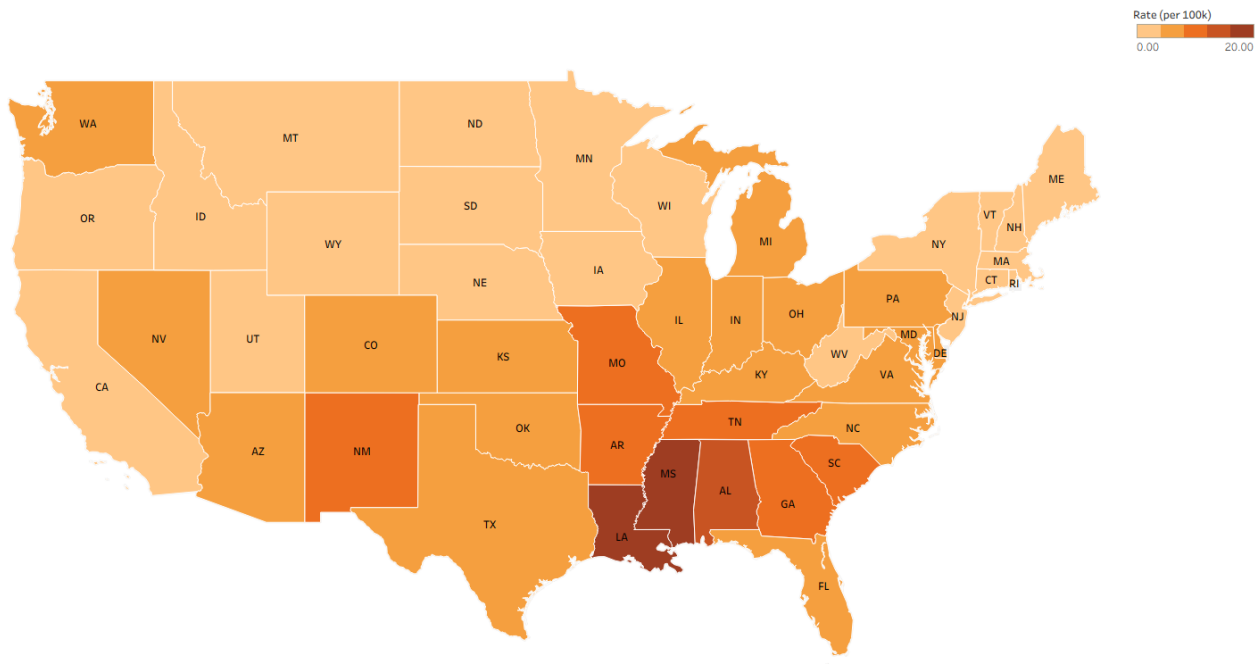
4 Visualizations

A consistent set of calculated fields and a shared color scheme were applied across all visualizations. Field names are self-explanatory (e.g. CF_law_count, CF_show_state, P_year) to ensure readability and reproducibility.

4.1 Q1 – Homicide Rates and the Law Landscape

Two charts answer Question 1^{2.6.1}. The first is a **choropleth map** displaying the firearm homicide rate across all U.S. states for a user-selected year, encoded with a sequential red palette so darker states indicate higher rates. The second is a **scatter plot** where each point represents one state: the X axis shows estimated gun ownership (FSS) as contextual background, the Y axis shows the composite law count, and color encodes the firearm homicide rate. A year parameter and a law-categorization parameter allow the viewer to scrub through time and switch the color encoding between law tier and law type group, enabling both a quantitative and a qualitative reading of the law landscape.

Firearm Homicide Rate by State (2023)



Map based on Longitude (generated) and Latitude (generated). Colour shows average of Firearm Homicide Rate. The marks are labelled by CF StateAbbreviation. Details are shown for CF LawGroupLabel and State. The data is filtered on Exclusions (Lawlevel,State) and Year. The Exclusions (Lawlevel,State) filter keeps 82 members. The Year filter has multiple members selected.

Figure 1: Q1 – Firearm Homicide Rate by State (Choropleth Map)

Gun Ownership vs Firearm Homicide Rate, by State (2023)

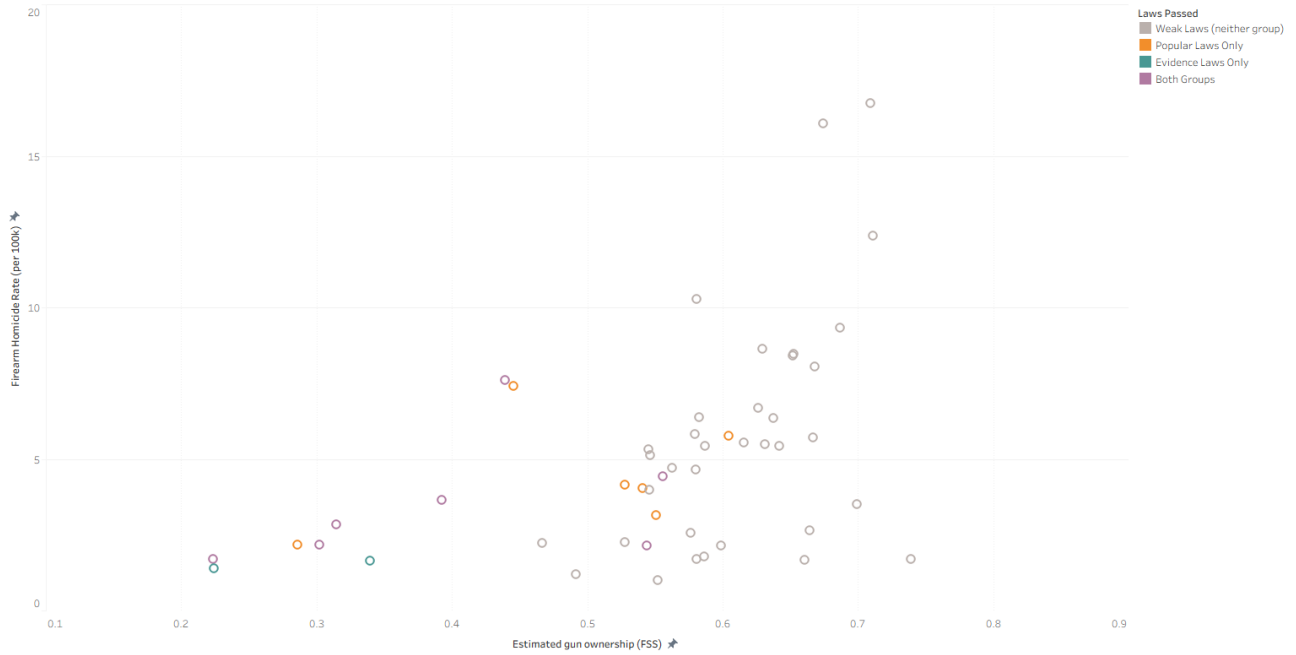


Figure 2: Q1 – Law Count vs. Estimated Gun Ownership (FSS), colored by Firearm Homicide Rate (Scatter Plot)

Insights: States with fewer laws cluster towards higher firearm homicide rates. The FSS axis provides ownership context but is not treated as a causal variable; the law count and law type dimensions are the primary analytical lenses.

4.2 Q2 – Homicide Trends by Law Profile Over Time

An **interactive line chart** answers Question 2^{2.6.2}. The X axis shows year (1976–2023), the Y axis shows firearm homicide rate, and each line represents a group of states defined by the active law-categorization parameter. When set to “Law Quantity Tier”, states are grouped into few-law, average-law, and high-law tertiles based on total law count. When set to “Law Type Group”, lines reflect groups defined by the categories of laws present. This toggle makes it possible to analyse both the volume and the nature of legislation in a single chart.

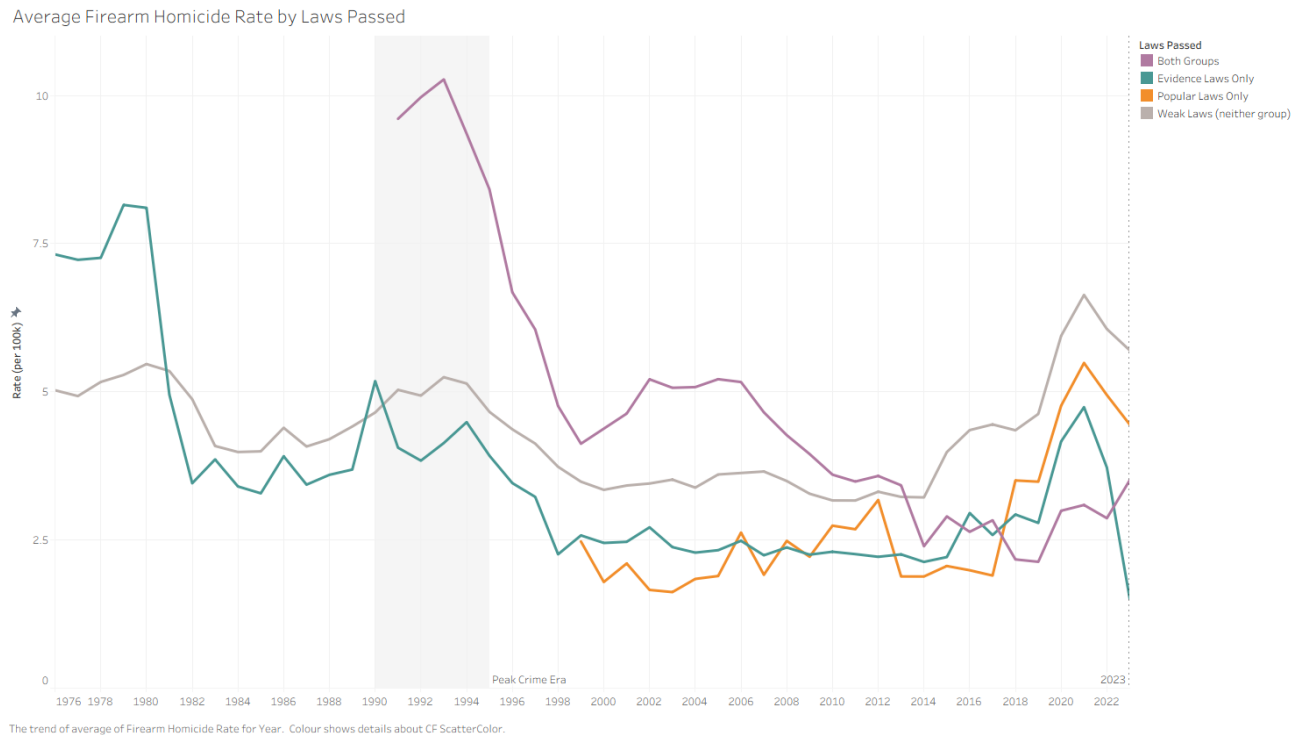


Figure 3: Q2 – Firearm Homicide Rate Over Time by Law Profile (1976–2023)

Insights: States grouped by higher law counts show a steeper decline in firearm homicide rates from the 1990s onward. The divergence between groups widens over time, and the pattern holds across both the law quantity and law type groupings.

4.3 Q3 – State-Level Law History and Homicide Standing

Two charts answer Question 3^{2.6.3}. The first is a **line chart** showing the total number of firearm laws in effect over time for a user-selected state (driven by the P_State parameter). This makes it easy to identify when a state underwent major legislative shifts. The second is a **bar chart** ranking all states by firearm homicide rate in a selected year: bars are colored by law-group category, and the user-selected state is highlighted so the viewer can immediately see where it stands relative to all others. Together, the two charts let the viewer judge whether periods of legislative growth in the selected state coincide with an improvement in its homicide ranking.

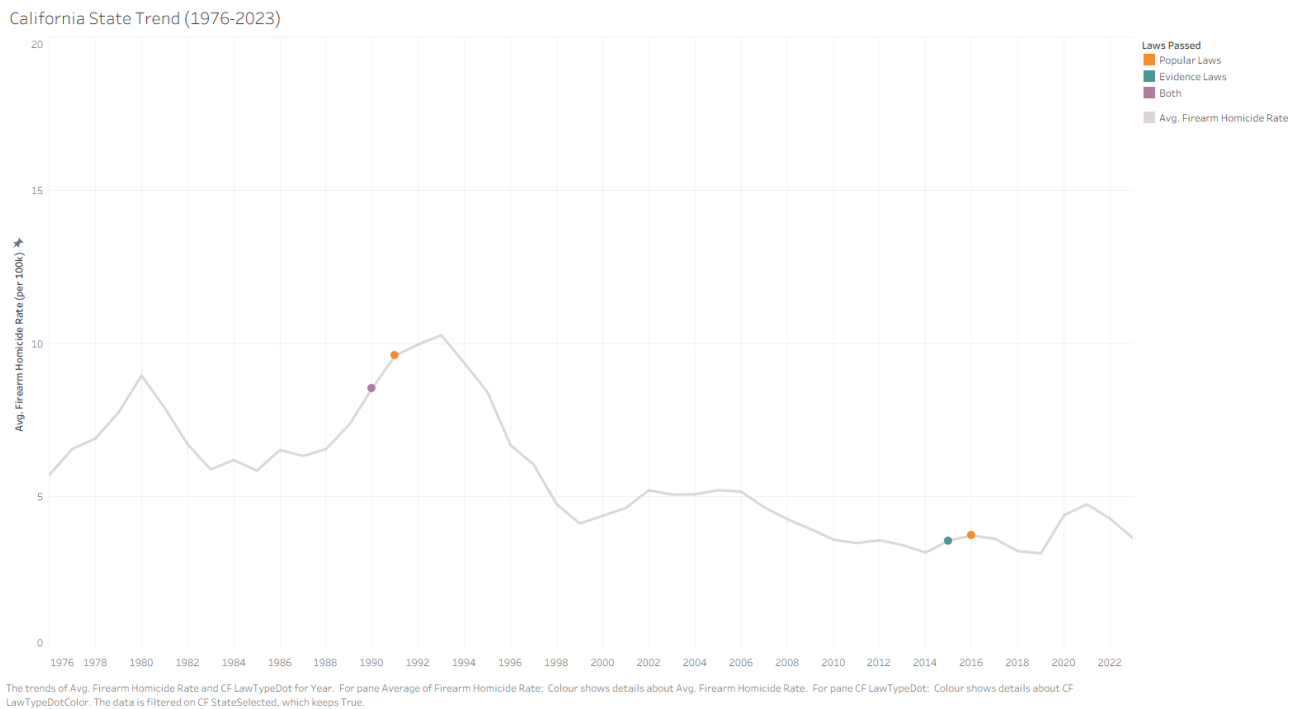


Figure 4: Q3 – Firearm Laws in Effect Over Time for Selected State (Line Chart)

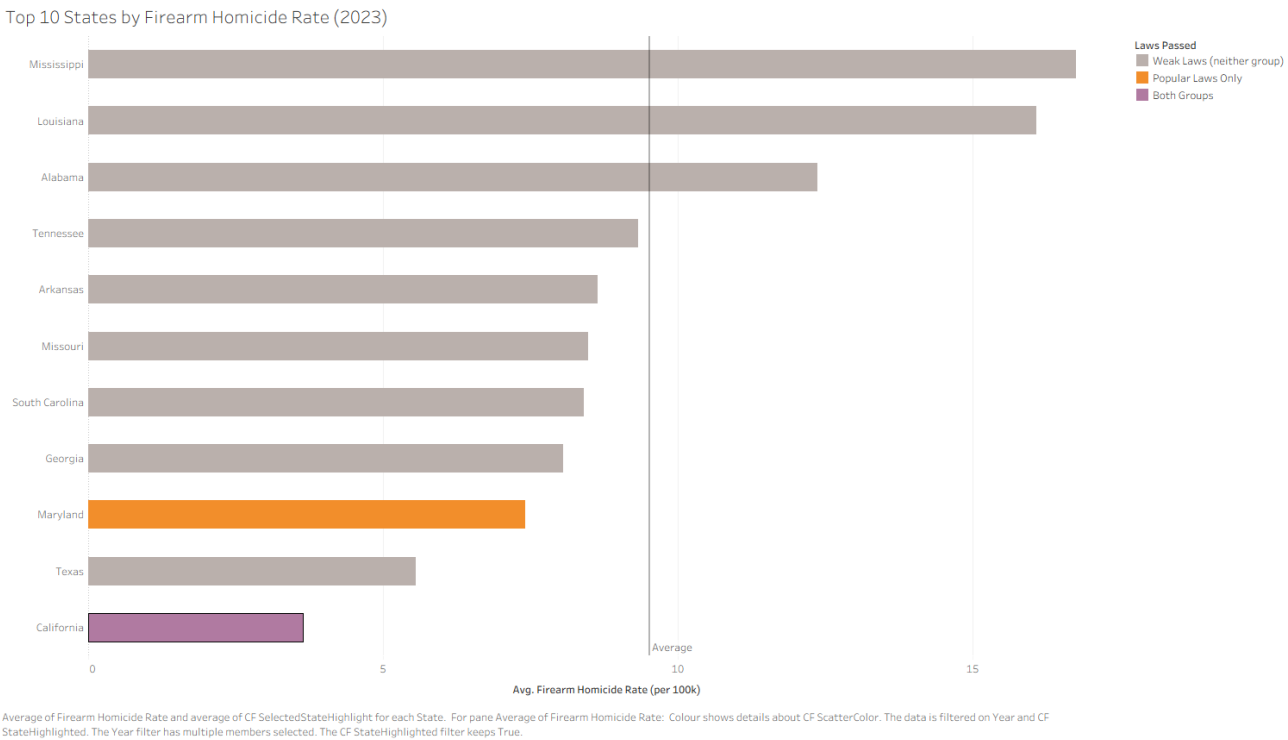


Figure 5: Q3 – Firearm Homicide Rate by State for Selected Year, colored by Law Group, Selected State Highlighted (Bar Chart)

Insights: In several states, sustained increases in law count coincide with a visible improvement in homicide ranking relative to other states. The relationship is not uniform and no causal interpretation is claimed; the viewer is invited to explore individual state trajectories interactively.

4.4 Dashboards

Two dashboards integrate the individual charts into a cohesive interactive experience.

Dashboard 1 combines the choropleth map (Q1Chart1), the scatter plot (Q1Chart2), and the law profile line chart (Q2Chart). A click action on the map filters the scatter to highlight the selected state; the law profile line chart provides immediate temporal context. The shared law-categorization parameter applies across all three views, meaning a single toggle switches the analytical lens for the entire dashboard. This gives the viewer a complete overview of the geographic, cross-sectional, and trend dimensions simultaneously.

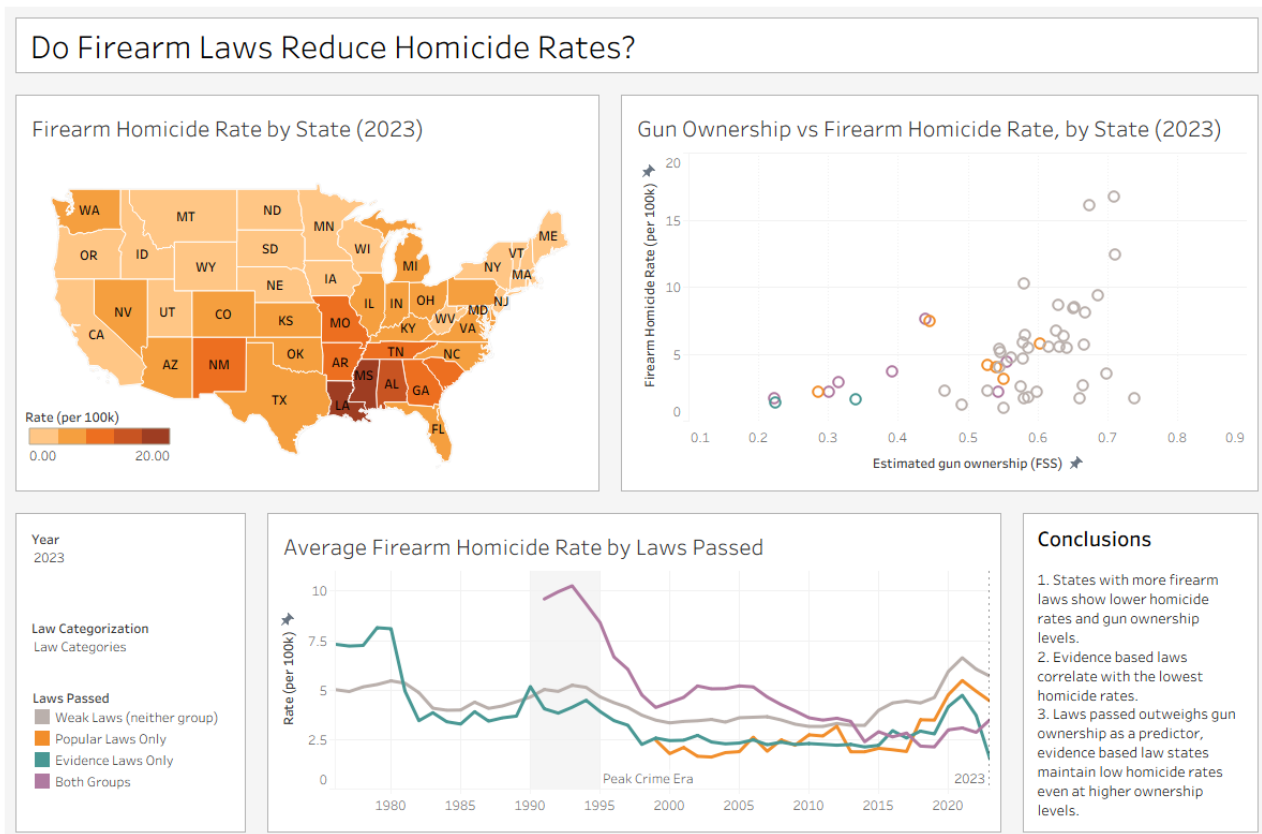


Figure 6: Dashboard 1 – Geographic Overview, Law–Homicide Scatter, and Law Profile Trends

Dashboard 2 combines the choropleth map (Q1Chart1), the comparative bar chart (Q3Chart2), and the single-state law history line chart (Q3Chart1). A shared state parameter drives both the bar chart highlight and the line chart, letting the viewer select any state and instantly see where it stands relative to all others and how its legislative history has unfolded over time.

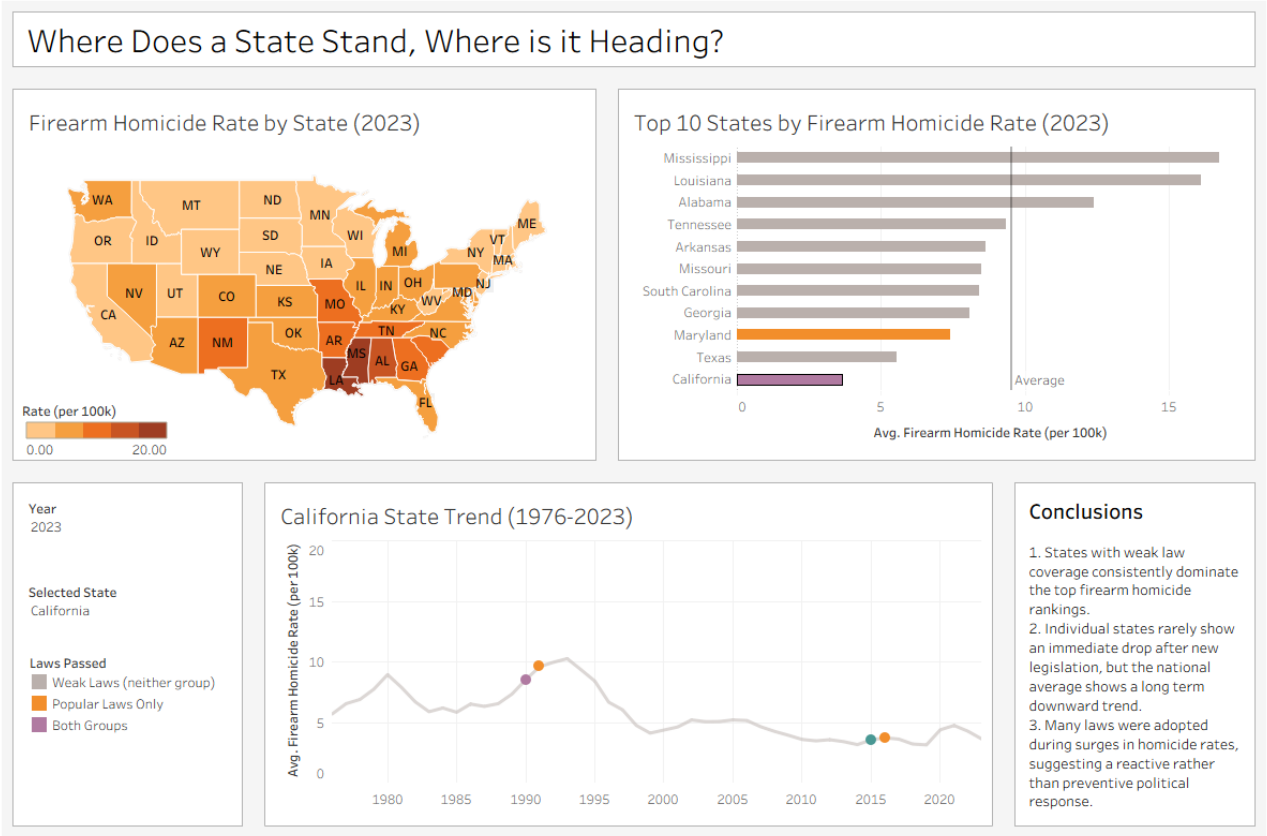


Figure 7: Dashboard 2 – Geographic Map, State Homicide Ranking Bar Chart, and Selected State Law History

4.5 Visualization Summary

Question	Chart	Chart Type	Tableau Feature
Q1 ^{2.6.1}	Fig. 1	Choropleth map	Calc. field, color encoding, param.
Q1 ^{2.6.1}	Fig. 2	Scatter plot	Parameter, calc. field, trend line
Q2 ^{2.6.2}	Fig. 3	Line chart (interactive)	Set, set action, reference line
Q3 ^{2.6.3}	Fig. 4	Line chart (per state)	Parameter, calc. field, filter
Q3 ^{2.6.3}	Fig. 5	Bar chart (comparative)	Calc. field, highlight, filter

Table 1: Visualization summary table

Table 1 summarizes:

- 4 chart types used across the project: choropleth map, scatter plot, line chart, and bar chart.
- 7 Tableau features used: calculated fields, parameters, sets, combined sets, dashboard set actions, reference lines, and trend lines.
- 2 data sources joined by state and year in Tableau, in two different file formats (.csv and .xlsx).

5 Conclusions

Visualizations reveal the following:

- States with fewer firearm laws consistently show higher firearm homicide rates, both geographically and in the cross-sectional scatter view [Fig. 2].
- States with more laws, whether measured by quantity or type category, have seen steeper declines in firearm homicide since the 1990s [Fig. 3].
- In several states, sustained legislative growth coincides with a visible improvement in homicide ranking relative to other states, though no causal claim is made [Fig. 4, Fig. 5].
- Gun ownership (FSS) provides useful contextual framing but is not treated as a causal variable; the regulatory environment is the primary analytical lens throughout the project.

Limitations include the proxy nature of the FSS ownership estimate, the absence of statistical controls for confounding variables such as poverty and urbanization, and the use of a static join on state name which may mask within-state variation.

6 References

- [1] Seokho Kang et al. *Firearm Suicide Proxy for Household Gun Ownership, 1949–2023*. Accessed: 2026-05-18. 2023. DOI: [10.7910/DVN/QVYDUD](https://doi.org/10.7910/DVN/QVYDUD). URL: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/QVYDUD>.
- [2] State Firearm Law Research Group. *State Firearm Law Database, 1976–2024*. Accessed: 2026-05-18. Tracks presence or absence of 72 firearm law provisions across all 50 U.S. states, 1976–2024. 2024. URL: <https://www.tuftsctsi.org/state-firearm-laws/>.
- [3] Seokho Kang and Taehee Lee. “State-level household gun ownership proxy dataset, 1949–2020”. In: *Data in Brief* 50 (2023), p. 109548. DOI: [10.1016/j.dib.2023.109548](https://doi.org/10.1016/j.dib.2023.109548). URL: <https://www.sciencedirect.com/science/article/pii/S2352340923006480>.
- [4] Seokho Kang et al. “Extending the Firearm Suicide Proxy for Household Gun Ownership”. In: *American Journal of Preventive Medicine* (2023). DOI: [10.1016/j.amepre.2023.06.004](https://doi.org/10.1016/j.amepre.2023.06.004). URL: <https://pubmed.ncbi.nlm.nih.gov/37160211/>.
- [5] Christopher S. Koper and Jeffrey A. Roth. *Impacts of the 1994 Assault Weapons Ban: 1994–96*. Tech. rep. NCJ 173405. Research in Brief. Congressionally mandated assessment of the Federal Assault Weapons Ban. National Institute of Justice, U.S. Department of Justice, 1999. URL: <https://www.ojp.gov/pdffiles1/173405.pdf>.
- [6] Christopher S. Koper. *An Updated Assessment of the Federal Assault Weapons Ban: Impacts on Gun Markets and Gun Violence, 1994–2003*. Tech. rep. Report to the National Institute of Justice. Jerry Lee Center of Criminology, University of Pennsylvania, 2004. URL: <https://www.ojp.gov/pdffiles1/nij/grants/204431.pdf>.
- [7] John J. Donohue. “The Effect of Permissive Gun Laws on Crime”. In: *The Annals of the American Academy of Political and Social Science* 704.1 (2022), pp. 115–138. DOI: [10.1177/00027162231164865](https://doi.org/10.1177/00027162231164865). URL: <https://journals.sagepub.com/doi/abs/10.1177/00027162231164865>.
- [8] Tableau Software. *Tableau Desktop*. 2026. URL: <https://www.tableau.com/products/desktop>.